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Lab 4

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04.1g: nginx Compute Engine

- **Take a screenshot of the site along with its Let's Encrypt certificate and include it in your lab notebook**





04.2g: Docker Guestbook

3. Build and run the Ubuntu-based container

- **Take a screenshot of the results for your lab notebook**

```
Successfully built 071ab47e9163
Successfully tagged helloubuntu:latest
evelynng@course-vm:~/cs430-src/04_container_dockerhub$ docker images

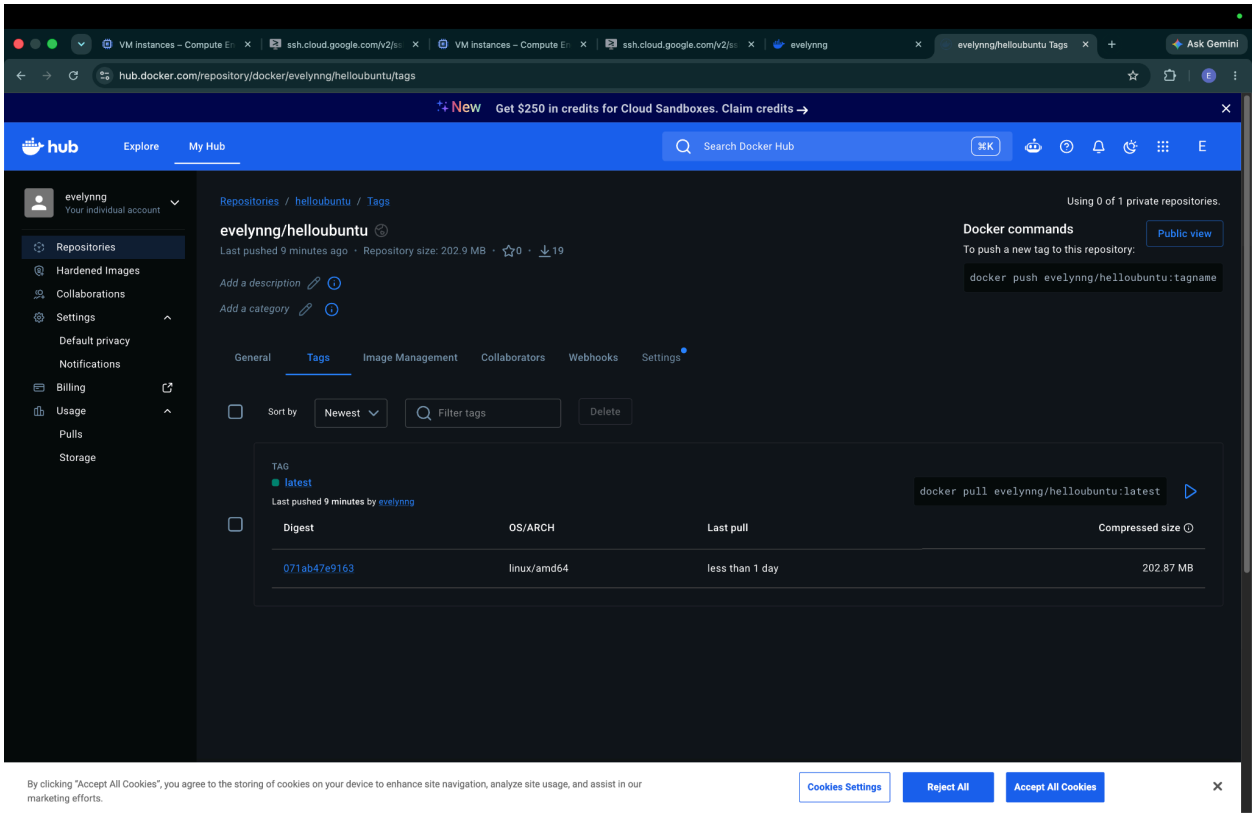
IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
helloubuntu:latest   071ab47e9163       762MB          213MB
ubuntu:22.04         0a9887931b0e       119MB         31.7MB
evelynng@course-vm:~/cs430-src/04_container_dockerhub$ evelynng
```

6. Running from Docker Hub

- **Run the image directly from Docker Hub and show a screenshot of the output of the command in your lab notebook.**

```
evelynng@course-vm:~/cs430-src/04_container_dockerhub$ docker run -di -p 8000:5000 --name hellou evelynng/helloubuntu
nable to find image 'evelynng/helloubuntu:latest' locally
atest: Pulling from evelynng/helloubuntu
4d3edee4da5: Pull complete
3301081a17b: Pull complete
c47da592aa1: Pull complete
7950225abe3: Pull complete
igest: sha256:071ab47e916380be6c69a39eb735d3738cdba1b794daf808f3f77de7032300b1
tatus: Downloaded newer image for evelynng/helloubuntu:latest
5c3213df1045ce864ce6ecfa26af82f33c3dfdb39d4c97fca996a9e9e2e3d0a
evelynng@course-vm:~/cs430-src/04_container_dockerhub$ evelynng
```

- **Take a screenshot of the container image and its size for your lab notebook.**



- **What layer adds the most to the container image? How much does it add?**

The `RUN apt-get install -y python3-pip` layer adds the most to the container image, contributing **374 MB**. This layer installs Python's package manager and its required dependencies.

8. Build and run the Alpine-based container

- **Take a screenshot of the image generated and its size for your lab notebook. How much smaller is the image than the Ubuntu one?**

```
Successfully built d8c576db713f
Successfully tagged helloalpine:latest
evelynng@course-vm:~/cs430-src/04_container_dockerhub$ docker images

IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
helloalpine:latest   d8c576db713f       104MB         25.2MB
python:alpine        d44b1254cb92      81.5MB         20.3MB
ubuntu:22.04         0a9887931b0e       119MB         31.7MB
evelynng@course-vm:~/cs430-src/04_container_dockerhub$ evelynng
```

The Alpine-based Docker image is 104 MB, while the Ubuntu-based Docker image is 762 MB, approximately 86.4% smaller than the Ubuntu image

- **Show the output of this command in a screenshot for your lab notebook. What might have happened?**

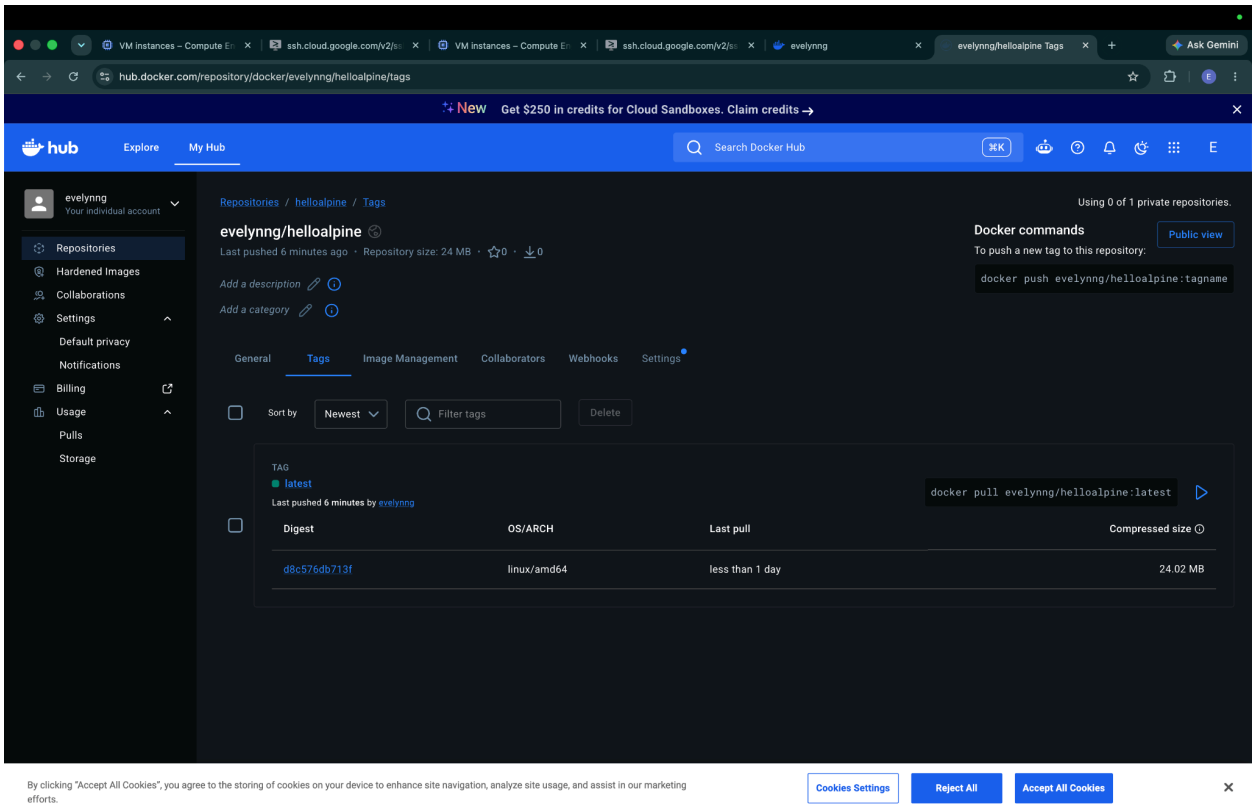
```
bad1a1b760ba   helloalpine    "python3 app.py"         47 seconds ago      Up 46 seconds                 0.0.0.0:8000->5000/tcp, [::]:8000->5000/tcp
ae3c3ff58a4a   f43032d4a80f   "/bin/sh -c 'apt-get…"   About an hour ago   Exited (0) About an hour ago
evelynng@course-vm:~/cs430-src/04_container_dockerhub$ docker exec -it helloa /bin/bash
OCI runtime exec failed: exec failed: unable to start container process: exec: "/bin/bash": stat /bin/bash: no such file or directory
evelynng@course-vm:~/cs430-src/04_container_dockerhub$ evelynng
```

- **Take a screenshot of the output of each**

```
OCI runtime exec failed: exec failed: unable to start container process: exec: "/bin/bash": stat /bin/bash: no such file or di
evelynng@course-vm:~/cs430-src/04_container_dockerhub$ docker exec -it helloa /bin/sh
/app # cat /etc/alpine-release
3.24.3
/app # ps -ef
PID   USER     TIME  COMMAND
    1 root      0:00 python3 app.py
    8 root      0:00 /usr/local/bin/python3 -c from multiprocessing.resource_tracker import main;main(5)
    9 root      0:00 /usr/local/bin/python3 app.py
   10 root      0:00 /usr/local/bin/python3 -c from multiprocessing.resource_tracker import main;main(6)
   20 root      0:00 /bin/sh
   27 root      0:00 ps -ef
/app #
```

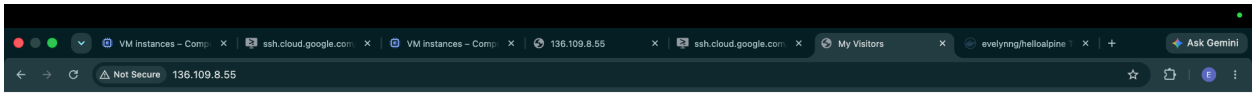
9. Docker Hub Alpine

- **Take a screenshot of the container image and its size.**



10. Compute Engine Ubuntu VM deployment

- **Take a screenshot of the entry that includes the VM's external IP address for your lab notebook**



Guestbook

Sign here

Entries

Evelyn Nguyen <evelynng@pdx.edu>
signed on 2026-10-26
Hello Compute Engine + Docker
